IBM Z DATATHON 2026 · SSF2026 · REAL-TIME AI FOR CRITICAL DECISIONS

Kavach

Stops UPI scams *inside* the transaction — by watching the payment, the money flow, and the scam call together.

In-transaction AI on IBM LinuxONE · Snap ML on the Telum on-chip AI accelerator

THE PROBLEM

Scam money leaves in seconds and is laundered in minutes

₹22,845 Cr

lost by Indians to cyber fraud in 2024 — a 206% jump over 2023.[1]

₹1,616 Cr

lost to "digital arrest" scams in just the first 9 months of 2024.[2]

24 lakh

Layer-1 mule accounts already reported to the national Suspect Registry.[1]

A victim on a fake "CBI" video call sends ₹50,000 over UPI. The receiving **mule** forwards it to the next account within minutes, then cashes out. By the time anyone reports it, the money is gone.

[1] MHA reply in Lok Sabha via Scroll.in (2025). [2] I4C data via The Indian Express / The Wire (2024).

WHY TODAY'S DEFENCES MISS IT

Each signal alone looks innocent

Rules

"Amount > ₹20k to a new payee" catches only **28%** of scams while blocking **2%** of honest payments — and scammers simply split amounts.

Per-transaction ML

Sees one payment at a time. It cannot see that the receiver collected from 9 strangers today, or that the sender just got off a scam call.

After-the-fact

Batch analytics and complaint portals act after settlement. Recovery after layering through mule chains is rare.

Stopping scams needs **more context** and a decision **before money settles** — on every payment, not a sample.

SOLUTION

Three signals, one decision, in microseconds

① The payment

Amount vs. the sender's habits, first payment to this person, new device, odd hour, bursts of payments.

② The money flow

Mule graph features: receiver fan-in from first-time payers, pass-through (money out minutes after money in). Repeat offenders go on a **mule watchlist** and are frozen.

③ The scam call

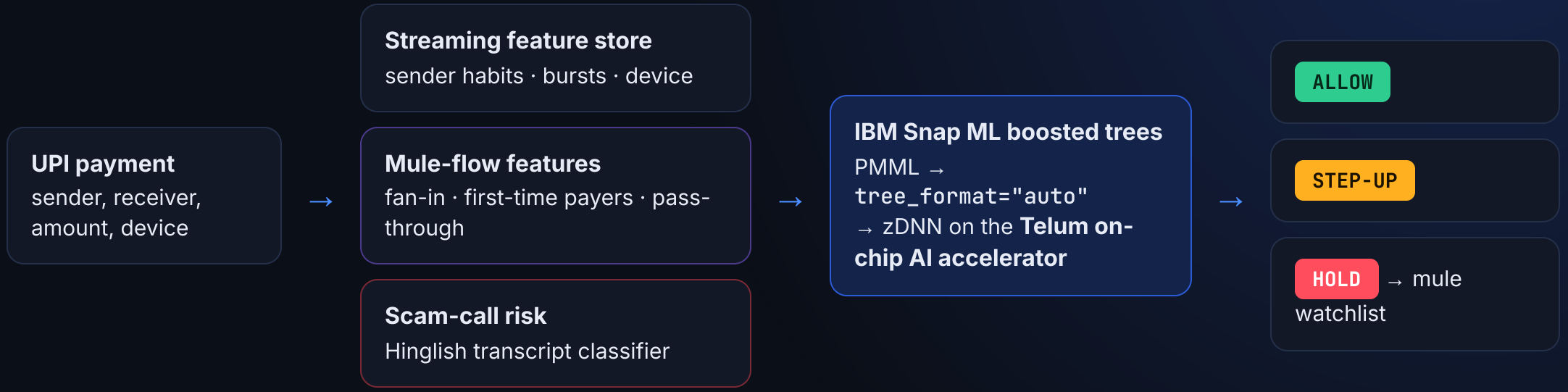
Hinglish call-transcript classifier spots "digital arrest", fake KYC, screen-share scripts. Risk attaches to the user for 2 hours.

ALLOW settle STEP-UP OTP + cooling-off + scam warning HOLD block, alert victim, notify bank

Every flag comes with plain-language reasons, e.g. "Scam call shortly before payment · Receiver collecting from many senders".

ARCHITECTURE

Scoring lives next to the payment, on IBM Z



One code path: the same streaming feature code builds training data and scores live payments, strictly from past events — no leakage, no training/serving skew. FastAPI + WebSocket dashboard; REST API for calls and payments.

WHY IBM Z / LINUXONE

Score 100% of payments in-line, where they already run

Telum on-chip AI accelerator

Snap ML compiles our trees to zDNN tensors on z16/z17, so inference runs on the processor chip next to the transaction workload — no network hop to a separate AI server.

Built and run on LinuxONE

Trained, benchmarked and served on a LinuxONE Community Cloud VM (s390x). One setup script; same model file falls back to CPU elsewhere.

Data never leaves the platform

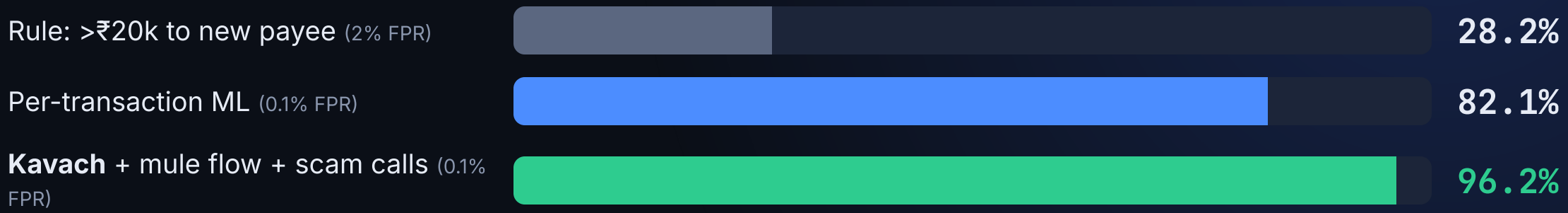
Payment and call data stay on Z with pervasive encryption. No external AI APIs, no data copies to a GPU cloud.

Scale & reliability

UPI does billions of payments a month with spikes at salary day and festivals. In-transaction AI must keep up with the core system itself — Z's throughput and uptime are made for this.

RESULTS · HELD-OUT DAYS, NEVER SEEN IN TRAINING

Same false-alarm budget, far more scams stopped



0.991
PR-AUC on 33,538 unseen payments (554 frauds)

26 µs
p50 full in-transaction path (features + model + decision), CPU baseline

0.06%
legit payments held in the live replay, while 97.8% of fraud payments were stopped

LIVE DEMO

Real-time dashboard: decisions, mule network, scam calls

Kavach
Real-time UPI scam & mule-network shield · in-transaction AI on IBM LinuxONE

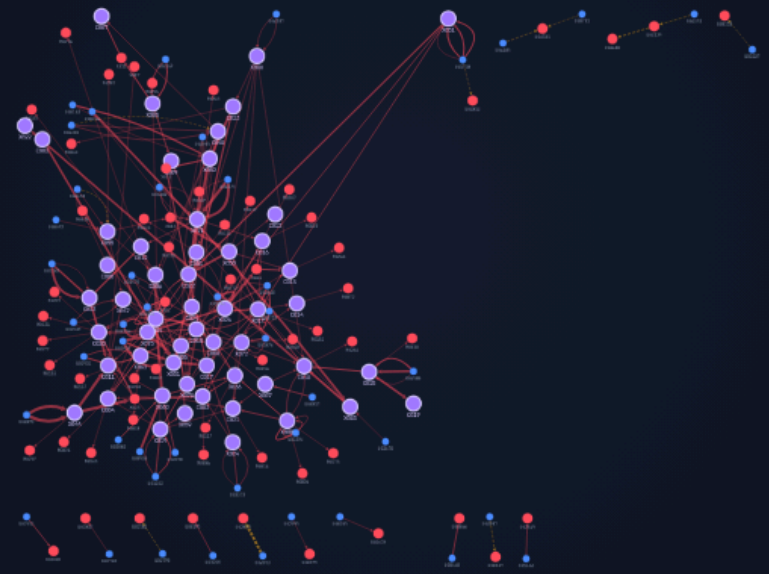
arch x86_64 | CPU compressed trees | Day 7 03:57 | Pause | 400 tx/s

SCORED	HELD	STEP-UP	MONEY PROTECTED	FRAUD CAUGHT	FALSE HOLDS	MULE WATCHLIST	LATENCY P50 / P99
17,393	377	13	₹49.9 L	97.0%	0.06%	60	103/249µs
UPI transactions	blocked before settlement	OTP + cooling-off	of ₹52.8 L scammed from victims	324 / 334 frauds	of legit payments	accounts frozen	features + inference

Live decisions — click a row for reasons · every flag + 1 in 5 allows shown

HOLD	05:10	X019 → M0076	₹890	mule acct
ALLOW	05:09	U01005 → M0081	₹1,289	legit
ALLOW	05:08	U03324 → M0106	₹151	legit
HOLD	05:07	U01016 → X075	₹18,447	ato
ALLOW	05:06	U00210 → M0225	₹512	legit
ALLOW	05:05	U02657 → M0043	₹935	legit
ALLOW	05:03	U00140 → M0105	₹978	legit
ALLOW	05:02	U00704 → M0232	₹1,770	legit
HOLD	05:02	U01016 → X075	₹43,587	ato
ALLOW	05:00	U01055 → U01494	₹1,500	legit
ALLOW	05:00	U01119 → M0111	₹121	legit
HOLD	04:59	U01016 → X075	₹56,357	ato
ALLOW	04:58	U04309 → M0116	₹2,201	legit
ALLOW	04:56	U01958 → U04991	₹4,532	legit

Mule network — victim · watchlisted mule · suspect receiver



DATA & HONESTY

Synthetic, but built to be hard

Real UPI data with mule labels is not public, so we simulate a reproducible UPI world: **5,000 users**, merchants, kirana shops, **90 mules + 25 cash-out accounts**, 7 days, **~117k payments, ~14k calls**.

Scams: digital arrest, fake KYC, account takeover, ₹1 "test" payments, repeat victims, multi-hop layering.

Hard negatives on purpose

Friends pooling trip money then paying an agent minutes later · legit big first-time payments · phone changes · night owls · real bank fraud-alert calls · older mule accounts · scam calls seen for only 55% of victims.

Limits: call classifier trained on template Hinglish; watchlist rule is simple. Everything is reproducible with one command.

BUILT FOR A PAYMENTS ENVIRONMENT

Secure by default, and it fails safe

Fail-safe

If scoring ever breaks, the payment is stepped up for verification, never silently allowed.

Tamper-evident model

The model file's SHA-256 is pinned at training time and checked at start-up. No pickle anywhere.

Audit trail

Every decision is logged with time, reasons and model version, ready for regulator and dispute review.

Locked-down API

API key with constant-time check, strict input schemas, per-IP rate limits, no CORS, docs off.

Hardened dashboard

Strict CSP, no inline scripts, SRI on CDN code, escaped output, anti-framing headers.

Abuse-tested

An attacker pushing dirty money into an innocent account cannot freeze it: the watchlist needs 2+ senders, expires, and only adds an OTP.

14 automated tests incl. auth, headers, rate limits, model tampering and abuse cases · pip-audit: 0 known vulnerabilities · server bound to localhost, viewed via SSH tunnel.

IMPACT & NEXT STEPS

From demo to a bank's payment switch

- Plugs into the bank / PSP authorisation path as a REST call — **no change for users** until risk is real.
- Cooling-off instead of hard blocks keeps honest users moving.
- Watchlist feeds the national **Suspect Registry**-style sharing between banks.
- Graph neural features over the full account graph.
- Federated training across banks without sharing raw data.
- On-device Hinglish ASR so call audio never leaves the phone.
- Benchmark at UPI scale on z17 with Spyre.

Kavach — the shield sits inside the payment, not after it.